

Effectiveness of sitagliptin and fixed dose combination (FDC) of Sitagliptin and Metformin in Indian Elderly patients with type 2 diabetes mellitus – Subset analysis of a Real-World Retrospective Study from Electronic Medical Records (EMR)



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Type 2 diabetes mellitus (T2DM) is a chronic metabolic disorder that is prevalent in India. The estimated prevalence of T2DM in **Indian adults is 11.4%**

In 2021, it was estimated that **101 million individuals in India** had diabetes

Older adults (age >65years) constitute nearly half of T2DM population

Presence of comorbidities, **increased predisposition to hypoglycaemic events**, and individual care needs leads to **increased risk of frailty, making disease management more complex** in older adults

Sitagliptin, and FDC of sitagliptin and metformin are effective in wide range of patients with T2DM, including elderly, and patients with comorbidities

This is a subset analysis of effectiveness of sitagliptin and sitagliptin and metformin FDC in elderly (Age >65 years) patients with T2DM

Real-world data are important to understand the generalizability and **applicability of sitagliptin and sitagliptin + Metformin FDC in clinical practice** and to inform clinicians about patient treatment decisions

To date, there is limited real-world data on effectiveness of sitagliptin and FDC of sitagliptin and Metformin in India

This **retrospective, observational, EMR-based study** aims to assess the real-world effectiveness and usage pattern of sitagliptin and FDC of Sitagliptin and metformin in Indian patients.

This is a subset analysis of proportion of elderly patients (Age $\geq$ 65yrs) achieving HbA1c <8% and HbA1c <7.5%

To evaluate the effectiveness of sitagliptin and fixed dose combination of Sitagliptin and Metformin in patients with type 2 diabetes mellitus in India

Study Design:

- This is a real-world, retrospective, observational, electronic medical record (EMR) based study
- In this study, EMR data from the patients satisfying the eligibility criteria was retrieved and analyzed

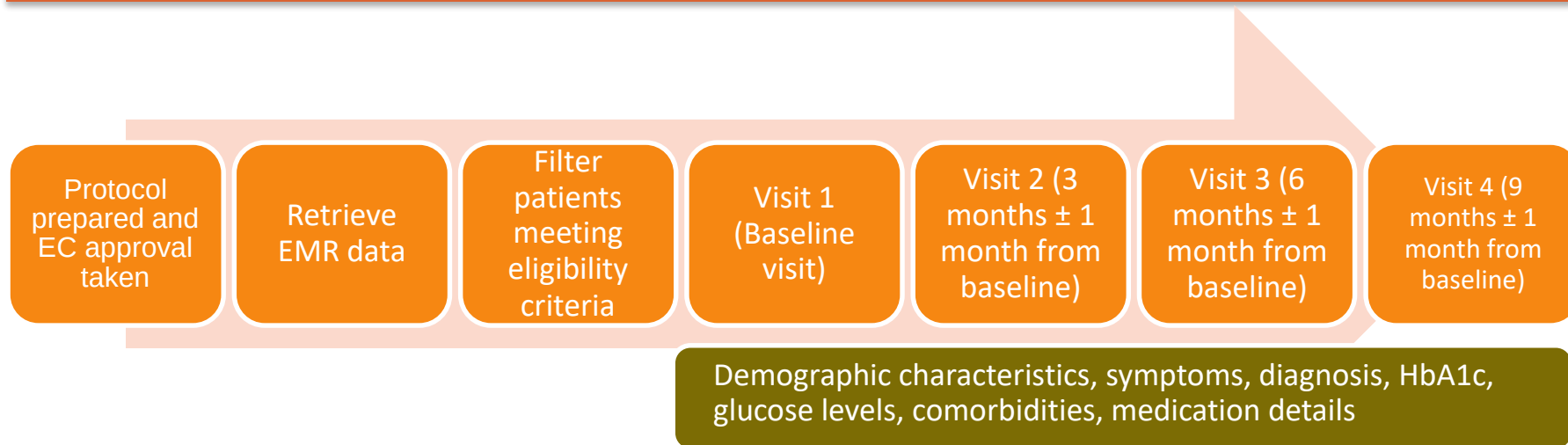
Inclusion Criteria:

- Male and female patients of age ≥ 18 years.
- Patients diagnosed with type-2 diabetes mellitus
- Patients with HbA1c $\geq 7\%$ at baseline visit
- Patients prescribed sitagliptin and FDC of sitagliptin and metformin in any visit other than baseline visit on the EMR platform
- Patients with EMR data captured for more than one visit

Exclusion Criteria:

- Patients diagnosed with type-1 diabetes mellitus.
- Patients are on insulin or any other injectable antidiabetic medication like glucagon-like peptide-1 (GLP -1) agonists
- Patients with incomplete medical records with respect to the required study parameters

The study retrieved EMRs of patients who met the eligibility criteria and had been prescribed sitagliptin and fixed-dose combination (FDC) of sitagliptin and metformin and were at least 18 years old.



- **The Baseline visit (Visit 1) reading was the visit in which the patient was prescribed with sitagliptin and fixed-dose combination (FDC) of sitagliptin and metformin as captured in EMR (2017-2023)**
- **And at follow-up visits (Visit 2, Visit 3 and Visit 4) reading after the baseline visit in this study.**

Primary Endpoint:

- Mean change in HbA1c from baseline to 3 months in patients with HbA1c $\geq 7\%$ at baseline.

Exploratory Endpoint:

1. Proportion of elderly patients (Age ≥ 65 yrs) achieving HbA1c $< 8\%$
2. Proportion of elderly patients (Age ≥ 65 yrs) achieving HbA1c $< 7.5\%$

Results: PATIENT FLOW

Patient Disposition (Category)	Patient count (n)
Total pool of patients on platform	2,98,24,372
Total patients in EMR records from 2017-2023 (with T2DM)	94,77,850
Total patients in EMR ≥ 18 years	86,21,919
EMR records of patients prescribed with sitagliptin alone or with other oral anti-diabetic medications	1,01,378
Patients prescribed with sitagliptin + metformin FDC alone or with other oral anti-diabetic medications	2,06,335
	Total patients
EMR records of patients prescribed with sitagliptin in any visit other than the baseline visit on EMR platform and have visits as per study time points (HbA1c $\geq 7\%$)	2,794
EMR records of patients prescribed with sitagliptin + metformin fixed-dose combination (FDC) in any visit other than the baseline visit on EMR platform and have visits as per the study time point (HbA1c $\geq 7\%$)	5,721
Sitagliptin	
Patients with EMR data captured (HbA1c) at baseline and 3 months (± 1 month)	1,008
Patients with EMR data (HbA1c) captured at baseline and 6 months (± 1 month)	585
Patients with EMR data (HbA1c) captured at baseline and 9 months (± 1 month)	302
Sitagliptin + Metformin FDC	
Patients with EMR data captured (HbA1c) at baseline and 3 months (± 1 month)	2,112
Patients with EMR data (HbA1c) captured at baseline and 6 months (± 1 month)	1,269
Patients with EMR data (HbA1c) captured at baseline and 9 months (± 1 month)	709

Demographic and Other Baseline Characteristics - Sitagliptin arm (N=2794)

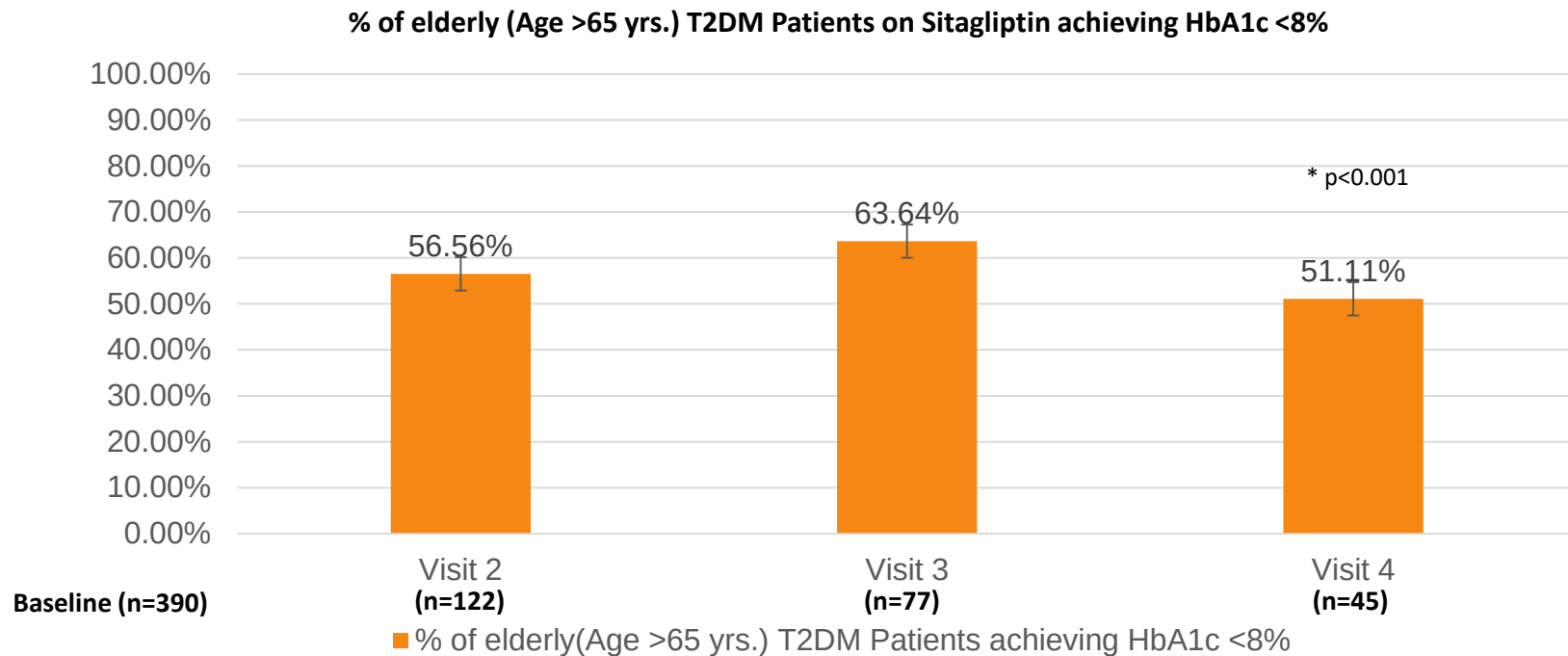
Parameter	Category	Patient Count(n)	Patient Percent(%)	Mean Value
Age (Years)	Overall	2794	100.00	55.68
	<=50	931	33.32	42.51
	51-60	862	30.85	55.54
	61-70	703	25.16	64.89
	>70	298	10.67	75.44
Gender	Male	1416	50.68	.
	Female	1378	49.32	.
Height (cm)	NA	1550	55.48	160.36
Weight (kg)	Overall	2210	100.00	69.65
	<60	479	21.67	53.75
	60-<80	1290	58.37	69.40
	80-<100	412	18.64	86.52
	100-<120	29	1.31	103.78
BMI (kg/m2)	Overall	1406	100.00	27.21
	=<25	466	33.14	22.75
	>25	940	66.86	29.42

Demographic and Other Baseline Characteristics - Sitagliptin + Metformin FDC arm (N= 5721)

Parameter	Category	Patient Count(n)	Patient Percent(%)	Mean Value
Age (Years)	Overall	5721	100.00	53.03
	<=50	2364	41.32	41.50
	51-60	1769	30.92	55.41
	61-70	1178	20.59	64.87
	>70	410	7.17	75.20
Gender	Male	3150	55.06	.
	Female	2571	44.94	.
Height (cm)	NA	3212	56.14	161.11
Weight (kg)	Overall	4411	100.00	72.41
	<60	663	15.03	54.60
	60-<80	2620	59.40	69.76
	80-<100	1004	22.76	87.17
	100-120	124	2.81	104.15
BMI (kg/m2)	Overall	2914	100.00	27.89
	=<25	783	26.87	22.91
	>25	2131	73.13	29.73

Results: Proportion of elderly patients (Age $\geq$ 65yrs)
achieving HbA1c <8% - Sitagliptin

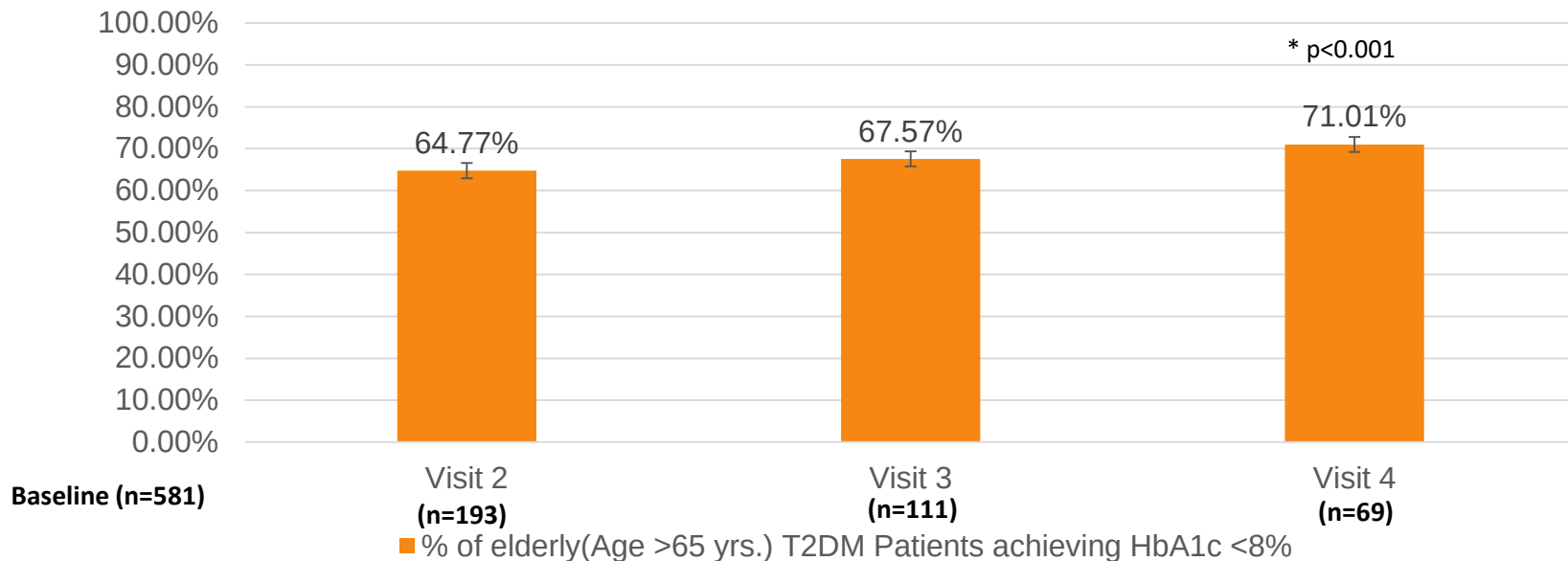
Results: % of elderly (Age >65 yrs.) T2DM Patients on Sitagliptin achieving HbA1c <8%



The mean HbA1c level at baseline was 9.45% in the sitagliptin arm. After a period of 3, 6 and 9 months, 56.56%, 63.64% and 51.11% elderly patients on sitagliptin achieved HbA1c of <8%.

Results: Proportion of elderly patients (Age $\geq$ 65yrs) achieving HbA1c $<8\%$ - FDC Sitagliptin and Metformin

% of elderly (Age >65 yrs.) T2DM Patients on Sitagliptin and Metformin FDC achieving HbA1c <8%

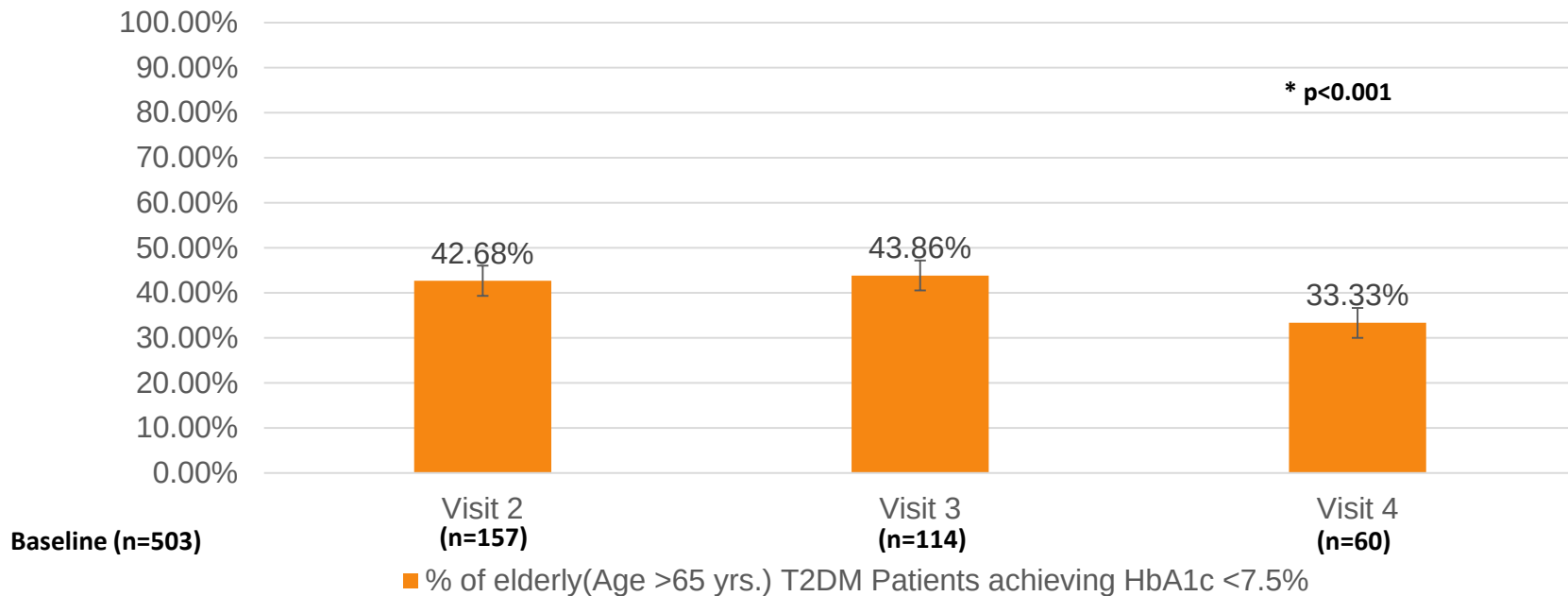


In these patients, the mean HbA1c level at baseline was 9.37% in the FDC of sitagliptin and metformin arm. In the subset of patients on sitagliptin and metformin FDC, 64.77%, 67.57% and 71.01% patients achieved HbA1c of <8% at 3, 6 and 9 months.

Results: Proportion of elderly patients (Age ≥ 65 yrs)
achieving HbA1c $< 7.5\%$ - Sitagliptin

Results: % of elderly(Age >65 yrs.) T2DM Patients on Sitagliptin achieving HbA1c <7.5%

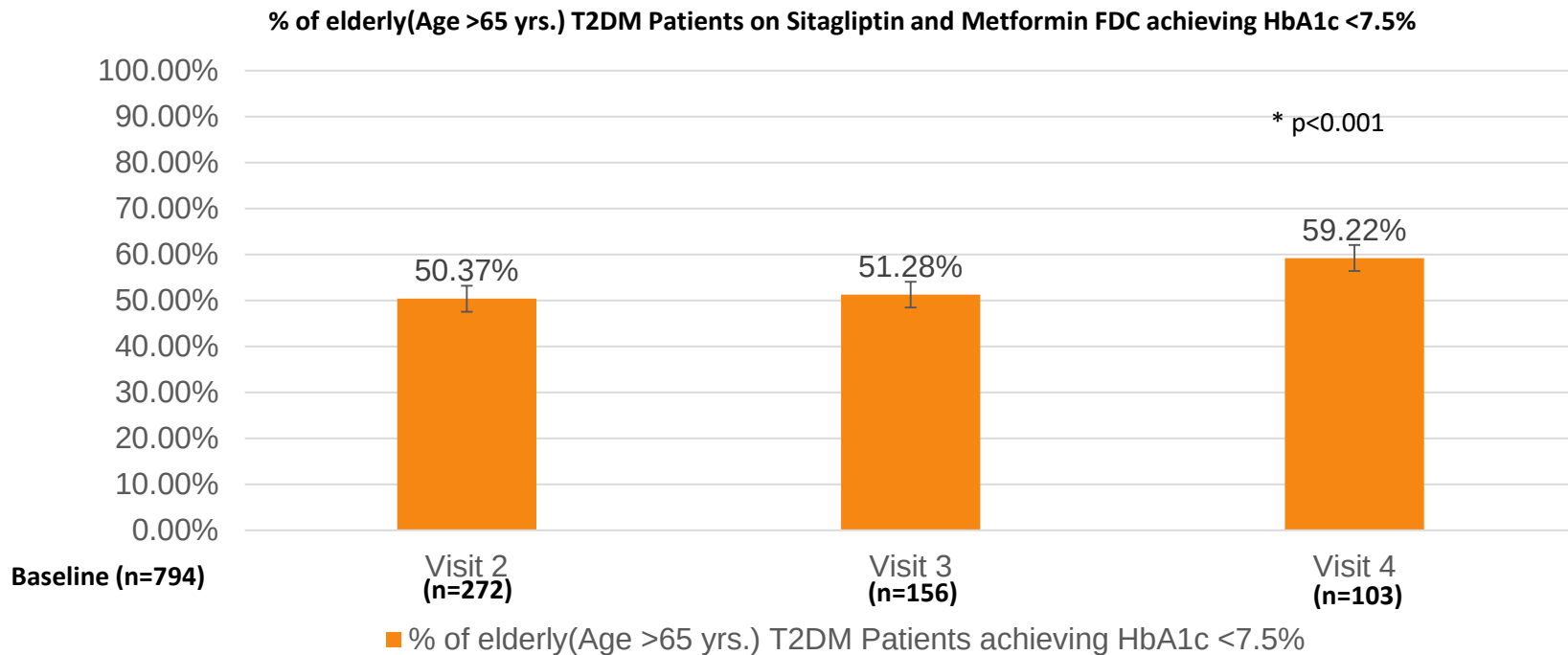
% of elderly(Age >65 yrs.) T2DM Patients on Sitagliptin achieving HbA1c <7.5%



In these patients, the mean HbA1c level at baseline was 9.06%. After a period of 3, 6 and 9 months, 42.68%, 43.86% and 33.33% elderly patients achieved HbA1c of <7.5%.

Results: Proportion of elderly patients (Age $\geq$ 65yrs)
achieving HbA1c $<7.5\%$ - FDC Sitagliptin and
Metformin

Results: % of elderly(Age >65 yrs.) T2DM Patients on Sitagliptin and Metformin FDC achieving HbA1c <7.5%



In these patients, the mean HbA1c level at baseline was 9.15%. After a period of 3, 6 and 9 months, 50.37%, 51.28% and 59.22% elderly patients achieved HbA1c of <7.5%.

- This a real world EMR based evidence and a **single arm analysis of the EMR data captured on digital platform**
- The data base on which this study was conducted, **safety data was not captured**

- This retrospective, EMR-based study demonstrated effectiveness of sitagliptin and sitagliptin and metformin FDC, in elderly patients with T2DM having HbA1c $\geq 7\%$, with significant proportion of patients achieving their HbA1c goals of $< 8\%$ and $< 7.5\%$.

- **ADA 2023: Elderly diabetics** with good quality of life with either none or very mild microvascular complications with a life expectancy of at least 10 to 15 years should have an A1C target of 7 -7.5%. **Those with diabetes with moderate cognitive impairments, microvascular complications, and comorbid conditions should have an A1C target of 7.5-8.5%**
- **RSSDI 2022: Elderly diabetics** with good quality of life with either none or very mild microvascular complications with a life expectancy of at least 10 to 15 years should have an A1C target of 7 -7.5%. **Those with diabetes with moderate cognitive impairments, microvascular complications, and comorbid conditions should have an A1C target of 7.5-8.5%**
- **RSSDI 2022:** In elderly patients with increased risk of hypoglycemia, use a **DPP-4 inhibitor as an alternative to sulfonylurea**. DPP-4 inhibitors may be a suitable addition to metformin for elderly patients (≥ 65 years) as there is very **low risk of hypoglycemia and weight gain; however, the dosage should be adjusted as per eGFR**. Recent RCTs have reported that gliptins are efficacious and safe with minimal side effects when used as add-on therapy in elderly patients with T2DM

Sitagliptin and FDC of sitagliptin and metformin represent a useful therapeutic option in elderly patients with T2DM for achieving glycemic control in real world setting and must be considered a part of comprehensive treatment plan

Thank you

- **Primary endpoint**
- Mean change in HbA1c from baseline to 3 months for patients with HbA1c $\geq 7\%$ at baseline in **Sitagliptin arm** – Total pool of patients:
 - -0.92% (n=1008)
- Mean change in HbA1c from baseline to 3 months for patients with HbA1c $\geq 7\%$ at baseline in **Sitagliptin + Metformin FDC arm** – Total pool of patients:
 - -1.03% (n=2113)